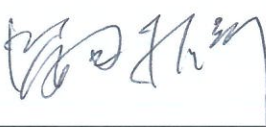


ISMS NETWORK MANAGEMENT

POLICY

Revision 00

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<u>Departemen Penanggung Jawab</u> <i>Responsible Department</i>	Information Technology Department

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INFORMATION SECURITY MANAGEMENT SYSTEM (ISMS)

NETWORK MANAGEMENT POLICY

1. PURPOSE

This policy sets the framework for managing and monitoring the organization's network infrastructure to ensure secure, reliable, and high-performance operations.

It defines the requirements for network management, including security measures, performance monitoring, and service level agreements (SLAs) to meet business continuity needs.

2. SCOPE

This policy applies to all network components and infrastructure, including routers, switches, firewalls, wireless access points, and other devices connected to the organization's network.

It applies to both internal staff and external contractors involved in the management of the network.

3. NETWORK MANAGEMENT RULES

This section outlines the procedures and standards for network management, including security and performance requirements.

3.1 Network Configuration and Access Control

- Network devices must be securely configured with the latest security patches and firmware.
- Access to network management consoles and configurations should be restricted to authorized personnel with multifactor authentication (MFA) enforced.
- All devices must be configured with role-based access control (RBAC) to ensure least-privilege access.

3.3 Network Segmentation

- Implement network segmentation to isolate sensitive systems and data from general user networks.
- Enforce strict access control lists (ACLs) and firewall rules to protect sensitive network segments from unauthorized access.

3.3 Monitoring and Logging

- All network devices and traffic must be monitored 24/7 using network performance monitoring tools.
- Intrusion detection and prevention systems (IDPS) must be implemented to detect suspicious or malicious activities.
- Network logs must be maintained and reviewed regularly to identify anomalies and incidents, and logs must be stored for at least 90 days.
- Employees must be informed that network activities are monitored for security and compliance purposes, and that usage of the company's network constitutes consent to such monitoring.

3.4 All unused ports must be:

- Assigned to an **isolated LAN** (VLAN 1 Default)
- Labeled clearly
- Set to **shutdown** or **disabled**

3.5. Network diagram

- All network need to be included in the network diagram (appendix 1)
- The network diagram (Appendix 1) needs to be reviewed annually and updated as the network changes.

4. NETWORK PERFORMANCE MONITORING

To ensure the availability and reliability of network services, Niterra Indonesia will deploy network performance monitoring tools and procedures.

These tools will provide real-time visibility into network operations and help meet performance SLAs.

4.1 Monitoring Tools

- Network Performance Monitoring tools must be used to track bandwidth usage, latency, jitter, and packet loss across critical network components.
- The network must be monitored for availability, traffic flow, and error rates across all key devices and connections.
- Alerts must be configured to notify administrators in case of performance degradation or network failures.

4.2 Performance Metrics

- Availability: Network uptime must meet the standards set out in the SLA (99.5% availability).
- Throughput: Minimum throughput for critical systems must be 100 Mbps, with higher throughput for core infrastructure.
- Latency: Latency between critical systems must not exceed 50 ms.
- Jitter: Jitter should not exceed 20 ms for real-time services like VoIP and video conferencing.
- Packet Loss: Packet loss should be kept below 0.1% on all connections.

5. SERVICE LEVEL AGREEMENT (SLA)

5.1 Network Availability

- The organization guarantees a 99.9% uptime for critical network infrastructure, excluding planned maintenance windows.
- Planned maintenance must be communicated to stakeholders at least 48 hours in advance and performed outside of business hours.

5.2 Response and Resolution Times

- Critical Incidents (e.g., total network outages or major security breaches) must be responded to within 30 minutes and resolved within 4 hours.
- High-Priority Issues (e.g., partial network failure or degraded performance) must be responded to within 1 hour and resolved within 8 hours.
- Low-Priority Issues (e.g., minor performance issues or isolated device failures) must be responded to within 4 hours and resolved within 48 hours.

5.3 Performance Guarantees

- Network throughput for core systems must meet or exceed 100 Mbps at all times.
- Latency between critical systems must not exceed 50 ms.
- Packet loss must remain below 0.1% across all network connections, ensuring high-quality performance for real-time applications like video and voice services.

5.4 SLA Monitoring and Review

- Network performance must be continuously monitored to ensure compliance with the SLA.
- The SLA and performance metrics will be reviewed annually, or as needed, to ensure they meet the changing needs of the business.

6. LAN & WLAN CONNECTION RULES

- Only authorized users are permitted to connect to the LAN & WLAN of Niterra Indonesia.
- Authorization must be granted by the IT department based on user roles and business needs.
- Role-Based Access Control (RBAC) must be enforced to limit network access to specific resources based on the user's job function.
- Users connecting to the LAN must authenticate using Windows Server DHCP list
- WLAN connections are provided by 3 conditions:
 - NITERRA-ID-Network : Specific for Niterra Employees
 - NITERRA-ID-Guest : Access connection using Vouchers that active for several times
 - NITERRA-ID-Devices : Specific Wireless connection for Niterra devices, such as : printers
- Voucher requests for guests must be made using the IT Services Request form at Niterra Portal.

7. LAN QUALIFICATIONS PROCESS

1. Device Qualifications

- Security Compliance: Devices must have up-to-date antivirus software, firewalls, and security patches installed. Regular vulnerability scans must be conducted to ensure compliance.
- Network Configuration: Devices must be properly configured to work within the Niterra DHCP configuration. Unauthorized software, services, or open ports that pose a security risk must be disabled.
- Device Registration: All devices must be registered with the IT department before they are allowed to access the LAN.

2. User Qualifications

All users must undergo a qualification process before they can connect to the LAN:

- Security Awareness Training: Users must complete security training that covers best practices for network access, password management, phishing prevention, and data protection.
- User Authentication: Users must authenticate using unique usernames and strong password.
- Access Rights Validation: The IT department must verify the user's access rights, ensuring that access to LAN resources is in line with the user's role and responsibilities.

8. NETWORK SEGMENTATION (VLAN)

Niterra Indonesia using VLAN to create segmentation for each department and machines.

The list are divided into:

VLAN	VLAN Name	Description
10	Server	Server Farm
20	IT	IT Segment

30	Finance	Finance Segment
40	Human_Capital	Human Capital Segment
50	Sales&Marketing	SM Segment
60	SCM	SCM Segment
70	FOS	FOS Segment
80	MTN&PE	MTN & PE Segment
87	Wifi_Guest	WLAN For Guest Segment
88	Wifi_User	WLAN For Employees
89	Wifi_Device	WLAN For Devices
90	Quality	QM Segment
99	Management	Master Management Segment
1000	Epsr-Control	Ring Control Connection

9. Network Services Risk Assessment

- 9.1 The IT Team must conduct a scheduled risk assessment of all network services at least every 6 months, including:
- VLAN and port configurations
 - Switch/router management access
 - SNMP, SSH, and remote access services
 - Unused or critical ports
- 9.3 Any identified vulnerability or misconfiguration must be resolved within **7 business days** of discovery.

10. BACKUP & RECOVERY

IT Department maintains regular backup of Firewall Configuration, the backup store in Server and tested regularly

- Network configurations must be backed up before making any changes to firmware.
- Configuration backups need to do at least annually.
- Recovery tests must be performed quarterly to ensure data can be restored in case of a disaster or major incident.

11. ROLES & RESPONSIBILITIES

7.1 Network Administrators

- Manage the day-to-day operations of the network, including performance monitoring, troubleshooting, and implementing security controls.
- Respond to network incidents and work towards resolving performance issues within SLA timelines.

7.2 IT Manager

- Oversee network security and performance, ensuring that monitoring tools and policies are enforced.
- Review performance reports and logs to identify issues and recommend improvements.

7.3 IT Support Staff

- Provide first-level support for network issues, escalating major issues to network administrators.
- Maintain network documentation, incident logs, and performance metrics.

7.4 Third-Party Vendors

- External vendors responsible for managing portions of the network must comply with the

organization's network management rules and SLA requirements.

- Vendors must report incidents and provide performance data to the organization's network team.

12. Risk Assessment Before Utilizing External Network Services

Before using any external network service (such as cloud services, VPNs, or third-party system integrations), a risk assessment must be conducted by the IT team. This ensures the service is secure, aligns with company policies, and does not pose risks to internal systems or data. The service may only be used after receiving approval from authorized personnel.

13. REVIEW & MAINTENANCE

This policy must be reviewed annually by the Information Security Manager and updated as necessary to reflect changes in technology, business needs, or regulatory requirements. The SLA must also be reviewed to ensure it continues to meet business performance requirements.

REVISION HISTORY				
No.	Date Revision	Rev	Pages / Points	Revision Material
01	September 2024	0	All	Create ISMS Network Management Policy
02	June 2025	1	Appendix 1	Update Network Diagram

Appendix 1 / Lampiran 1: Network Diagram

